MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – TEST (SET - B)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 25**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

MATHS MANTHAN | $(5 \times 1 = 5$ Marks)

Q1. The ratio of the length of a pole and its shadow on the ground is $1 : \sqrt{3}$. The angle of elevation of the Sun is: [1]

(A) $90°$ (B) $60°$ (C) $45°$ (D) $30°$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D)

Justification: $\tan\theta = \dfrac{1}{\sqrt{3}} = \tan 30° \implies \theta = \boxed{30°}$.

Q2. A vertical pole 10 m long casts a shadow of length 5 m on the ground. At the same time, a tower casts a shadow of length 12.5 m on the ground. The height of the tower is: [1]

(A) 25 m (B) 20 m (C) 15 m (D) 10 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: $\dfrac{H}{12.5} = \dfrac{10}{5} = 2 \implies H = \boxed{25 \text{ m}}$.

Q3. A kite is flying at a height of 150 m from the ground. It is attached to a string inclined at an angle of $30°$ to the horizontal. The length of the string is: [1]

(A) 150 m (B) $150\sqrt{3}$ m (C) 300 m (D) $300\sqrt{3}$ m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

Justification: $\sin 30° = \dfrac{150}{L} \implies L = \dfrac{150}{1/2} = \boxed{300 \text{ m}}$.

Q4. An observer 1.8 m tall stands away from a chimney at a distance of 38.2 m along the ground. The angle of elevation of the top of the chimney from his eyes is 45°. The height of the chimney is: [1]

(A) 40 m (B) 38.2 m (C) 36.4 m (D) 41.8 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: Height $= 1.8 + 38.2 \tan 45° = 1.8 + 38.2 = \boxed{40 \text{ m}}$.

Q5. A wire is attached from a point A on the ground to the top of a pole BC, making an angle of elevation of 60°. If $AB = 5\sqrt{3}$ m, then the length of the wire is: [1]

(A) 10 m (B) $10\sqrt{3}$ m (C) 15 m (D) $5\sqrt{3}$ m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)

Justification: In right $\triangle ABC$ ($\angle B = 90°$), $AB = 5\sqrt{3}$ m is the horizontal base and AC is the wire:

$\cos 60° = \dfrac{AB}{AC} \implies AC = \dfrac{5\sqrt{3}}{1/2} = \boxed{10\sqrt{3} \text{ m}}$.

SECTION B: Very Short Answer Questions

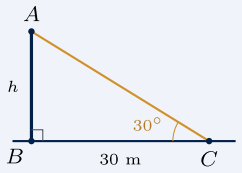
MATHS MANTHAN | ($2 \times 2 = 4$ Marks)

Q6. The angle of elevation of the top of a tower from a point on the ground which is 30 m away from the foot of the tower, is 30°. Find the height of the tower. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let tower $AB = h$ m, distance $BC = 30$ m, $\angle ACB = 30°$.

[0.5 Mark]



$\tan 30° = \dfrac{h}{30} \implies h = \dfrac{30}{\sqrt{3}} = 10\sqrt{3} = \boxed{10\sqrt{3} \text{ m} \approx 17.32 \text{ m}}$

Height computation: [1.5 Marks]

Q7. A 1.5 m tall boy is walking away from the base of a lamp post which is 12 m high, at the speed of 2.5 m/sec. Find the length of his shadow after 4 seconds. [2]

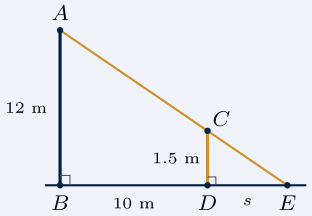
▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let lamp post $AB = 12$ m and boy $CD = 1.5$ m walk away from base B at 2.5 m/s for 4 s, so $BD = 2.5 \times 4 = 10$ m. Let shadow $DE = s$ m.

[0.5 Mark]

Since $\triangle ABE \sim \triangle CDE$ (AA similarity): $\dfrac{AB}{CD} = \dfrac{BE}{DE} \implies \dfrac{12}{1.5} = \dfrac{10+s}{s}$.

[1 Mark]



$\implies 8s = 10 + s \implies 7s = 10 \implies s = \boxed{\dfrac{10}{7} \text{ m} \approx 1.43 \text{ m}}$

Solving for shadow s: [0.5 Mark]

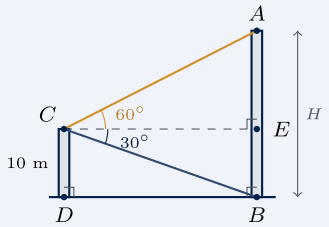
Q8. A man standing on the deck of a ship, which is 10 m above water level, observes that the angle of elevation of the top of a hill is 60° and the angle of depression of the base of the hill is 30°. Find the distance of the hill from the ship and the height of the hill. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Let deck point C be at $CD = 10$ m above water level and hill be AB. Draw $CE \perp AB$ so $EB = 10$ m, $\angle ACE = 60°$, $\angle CBD = 30°$. [0.5 Mark]

In $\triangle CDB$: $BD = CE = 10\cot 30° = 10\sqrt{3} \approx 17.32$ m. [1 Mark]

In $\triangle AEC$: $AE = 10\sqrt{3}\tan 60° = 30$ m. [1 Mark]



Height of hill $= AB = AE + EB = 30 + 10 = 40$ m.

$\therefore$ Distance of hill $= \boxed{10\sqrt{3} \text{ m} \approx 17.32 \text{ m}}$, Height of hill $= \boxed{40 \text{ m}}$

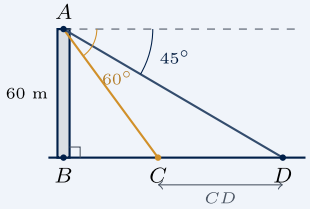
Final values: [0.5 Mark]

Q9. As observed from the top of a lighthouse 60 m high from sea level, the angles of depression of two ships are 45° and 60°. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships. (Use $\sqrt{3} = 1.732$) **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Let lighthouse $AB = 60$ m, ships on same side be C (60°) and D (45°). [0.5 Mark]

In $\triangle ABC$: $BC = \dfrac{60}{\tan 60°} = 20\sqrt{3}$ m. In $\triangle ABD$: $BD = 60\cot 45° = 60$ m. [2 Marks]



Distance between ships $= CD = 60 - 20\sqrt{3} = 60 - 20(1.732) = 60 - 34.64 = \boxed{25.36 \text{ m}}$. [0.5 Mark]

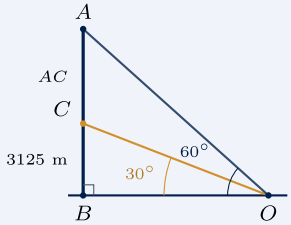
Q10. An aeroplane when flying at a height of 3125 m from the ground passes vertically below another plane at an instant when the angles of elevation of the two planes from the same point on the ground are 30° and 60° respectively. Find the distance between the two planes at that instant. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Let lower plane C ($CB = 3125$ m) pass directly below upper plane A at the same vertical line AB, viewed from O on the ground ($\angle COB = 30°, \angle AOB = 60°$). [0.5 Mark]

In right $\triangle CBO$: $\tan 30° = \dfrac{3125}{OB} \implies OB = 3125\sqrt{3}$ m. [1 Mark]

In right $\triangle ABO$: $\tan 60° = \dfrac{AB}{OB} \implies AB = (3125\sqrt{3})\sqrt{3} = 9375$ m. [1 Mark]



Vertical separation between the two planes $= AC = AB - CB = 9375 - 3125 = \boxed{6250 \text{ m}}$. [0.5 Mark]

Note: The source question paper header specifies MM: 25, while the printed section breakdown totals $5+4+9+5=23$ Marks.

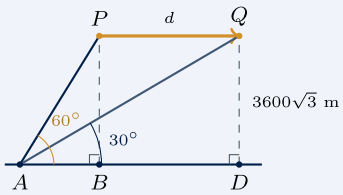
Q11. The angle of elevation of an aircraft from a point A on the ground is $60°$. After a flight of 30 seconds, the angle of elevation changes to $30°$. The aircraft is flying at a constant height of $3600\sqrt{3}$ m. Find the speed of the aircraft in km/h. **[5]**

OR

A person standing on the bank of a river observes that the angle of elevation of the top of a tower on the opposite bank is $60°$. When he moves 30 m away from the bank, he finds the angle of elevation to be $30°$. Find the height of the tower and the width of the river. (Take $\sqrt{3}=1.732$) **[5]**

▶ MAIN QUESTION — STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let aircraft fly horizontally from P to Q (above ground points B, D) at constant height $3600\sqrt{3}$ m, observed from A at $60°$ and $30°$. [1 Mark]



In $\triangle PBA$: $AB = \dfrac{3600\sqrt{3}}{\tan 60°} = 3600$ m. In $\triangle QDA$: $AD = \dfrac{3600\sqrt{3}}{\tan 30°} = 3600 \times 3 = 10800$ m. [2 Marks]

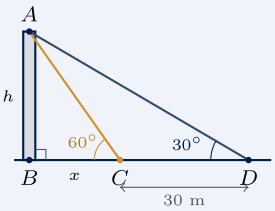
Distance covered in 30 s: $BD = 10800 - 3600 = 7200$ m. [1 Mark]

$$\text{Speed} = \frac{7200}{30} = 240 \text{ m/s} = 240 \times \frac{18}{5} = \boxed{864 \text{ km/h}}$$

Final speed in km/h: [1 Mark]

▶ OR CHOICE — STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let tower $AB = h$ m, river width $BC = x$ m, and second observation point D have $CD = 30$ m, with $\angle ACB = 60°, \angle ADB = 30°$. [1 Mark]



In $\triangle ABC$: $h = \sqrt{3}\,x$ — (i); in $\triangle ABD$: $h = \dfrac{x+30}{\sqrt{3}}$ — (ii). [1.5 Marks]

Equating (i) and (ii): $\sqrt{3}\,x = \dfrac{x+30}{\sqrt{3}} \implies 3x = x+30 \implies 2x = 30 \implies x = 15$ m. [1.5 Marks]

Height $h = 15\sqrt{3} = 15(1.732) = 25.98$ m.

$$\therefore \text{ Width of river} = \boxed{15 \text{ m}}, \qquad \text{Height of tower} = \boxed{15\sqrt{3} \text{ m} \approx 25.98 \text{ m}}$$

Both required values: [1 Mark]

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★